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Contents

Theory (Logic)	2
Pseudo Code	3
Program Code	4
Output	6
Conclusion	7

Theory (Logic)

This is a C++ program that shows how to convert a distance from centimeters to meters using classes. There are two classes: Centimeter and Meter. The Centimeter class stores a distance in centimeters, and the Meter class stores a distance in meters. In the main() function, we first create an object of the Centimeter class and store a value (250 cm). Then, we convert this object into a Meter object. This conversion happens automatically using a constructor in the Meter class that accepts a Centimeter object. Inside that constructor, it takes the centimeter value and divides it by 100 to get meters. Then, both values are printed on the screen.

Pseudo Code

- **Step 1: Include <iostream>**
- **Step 2: Use the standard namespace**
- **Step 3: Create a class named Centimeter**
 - **Private:**
 - **centimeters**
 - **Create a constructor: takes centimeter value**
 - **Function: return centimeter value**
 - **Function: display value in centimeters**
- **Step 4: Create class named Meter**
 - **private**
 - **meters**
 - **constructor: set meter value to 0**
 - **constructor**
 - **takes Centimeter object**
 - **convert the cm to meters by dividing by 100**
- **Step 5: Start main function**
 - **Centimeter object with value 250**
 - **Print the centimeter value**
 - **Meter object from the Centimeter object**
 - **Print the meter value**
- **Step 6: End main function**

Program Code

```
#include <iostream>
using namespace std;

class Centimeter
{
    private:
        float cm;

    public:
        Centimeter(float c)
        {
            cm=c;
        }

        float getCm() const
        {
            return cm;
        }

        void display() const
        {
            cout<<"distance in cm: "<<cm<<"cm"<<endl;
        }
};

class Meter
{
    private:
        float meter;

    public:
        Meter()
```

```

{
    meter=0;
}

Meter(const Centimeter& c)
{
    meter=c.getCm()/100.0;
}

void display() const
{
    cout<<"distance in meters: "<<meter<<"m"<<endl;
}
};

int main()
{
    Centimeter cm(250);
    cm.display();

    Meter m=cm;
    m.display();

    return 0;
}

```

Output

```
distance in cm: 250cm  
distance in meters: 2.5m
```

Conclusion

Code is run successfully, required output is received.